

Paper Completion Checklist

Representations of Lie Groups: A Mathematical Dissection of Quantum Spin

Generated July 20, 2026

11-Day Sprint: Jul 20–31 (Presentation on Fri 31)											
20	21	22	23	24	25	26	27	28	29	30	31
Mon	Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri
D1	D2	D3	D4	D5	D6	D7	D8	D9	Buffer	Review	PRES.

Mon 20 Jul — Day 1: $\mathfrak{so}(3)$ & $\mathfrak{su}(2)$ Lie Algebras

- ☐ Define $\mathfrak{so}(3) = T_{\mathbf{I}_3}\mathrm{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3,3} \mid \mathbf{R}^\top = -\mathbf{R}\}$
- ☐ Define basis $\mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3$ with explicit matrices
- ☐ Prove $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk}\mathbf{L}_k$ and $\dim \mathfrak{so}(3) = 3$
- ☐ Define $\mathfrak{su}(2) = T_{\mathbf{I}_2}\mathrm{SU}(2) = \{\mathbf{U} \in \mathbb{C}^{2,2} \mid \mathbf{U}^\dagger = -\mathbf{U}, \mathrm{tr}(\mathbf{U}) = 0\}$
- ☐ Define basis $\mathbf{T}_i = \frac{1}{2i}\sigma_i$ with explicit Pauli matrices
- ☐ Prove $[\mathbf{T}_i, \mathbf{T}_j] = \varepsilon_{ijk}\mathbf{T}_k$ and $\dim \mathfrak{su}(2) = 3$

Tue 21 Jul — Day 2: Double Cover & Lie Algebra Isomorphism

- ☐ Define $\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ by $\Phi(\mathbf{U})(\mathbf{X}) = \mathbf{U}\mathbf{X}\mathbf{U}^\dagger$
- ☐ Prove $\Phi(\mathbf{U}) \in \mathrm{SO}(3)$ (preserves inner product and determinant)
- ☐ Prove Φ is a smooth group homomorphism
- ☐ Prove Φ is surjective with $\ker \Phi = \{\pm \mathbf{I}_2\}$ (2-to-1)
- ☐ Conclude $\mathrm{SU}(2)$ is the universal double cover of $\mathrm{SO}(3)$
- ☐ Define $\phi : \mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$ by $\phi(\mathbf{L}_i) = \mathbf{T}_i$
- ☐ Prove ϕ is a Lie algebra isomorphism (linear + preserves bracket)

Wed 22 Jul — Day 3: Complexification & Ladder Operators

- ☐ Define complexification: $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ for a real Lie algebra $\mathfrak{g}$
- ☐ Define complex representation of a real Lie algebra (extends to $\mathfrak{g}_{\mathbb{C}}$)
- ☐ Prove $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C}) = \{\mathbf{A} \in \mathbb{C}^{2,2} \mid \mathrm{tr}(\mathbf{A}) = 0\}$
- ☐ Define Cartan–Weyl basis: $\mathbf{H} = 2i\mathbf{T}_3, \mathbf{E} = i\mathbf{T}_1 - \mathbf{T}_2, \mathbf{F} = i\mathbf{T}_1 + \mathbf{T}_2$
- ☐ Write explicit matrix forms of $\mathbf{H}, \mathbf{E}, \mathbf{F}$
- ☐ Prove commutation relations: $[\mathbf{H}, \mathbf{E}] = 2\mathbf{E}, [\mathbf{H}, \mathbf{F}] = -2\mathbf{F}, [\mathbf{E}, \mathbf{F}] = \mathbf{H}$

Thu 23 – Fri 24 Jul — Days 4–5: Classification of Irreducible Representations

- ☐ State the theorem: for each $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, $\exists!$ irrep $\pi_j : \mathfrak{su}(2) \rightarrow \mathfrak{gl}(V_j)$ with $\dim V_j = 2j+1$
- ☐ Define weight spaces and prove $\pi_j(\mathbf{H})$ diagonalizes on V_j
- ☐ Ladder Lemma: $\pi_j(\mathbf{E})$ raises weight by $+2$, $\pi_j(\mathbf{F})$ lowers by -2

- ☐ Highest weight argument: pick $\mathbf{v} \neq \mathbf{0}$ with $\pi_j(\mathbf{E})\mathbf{v} = \mathbf{0}$, let λ be its $\mathbf{H}$ -eigenvalue
- ☐ Construct basis $\mathbf{v}_k = \pi_j(\mathbf{F})^k \mathbf{v}$ and prove $\pi_j(\mathbf{E})\mathbf{v}_k = k(\lambda - k + 1)\mathbf{v}_{k-1}$
- ☐ Prove $\lambda = 2j \in \mathbb{N}_0$ and the chain terminates at $\mathbf{v}_{2j+1} = \mathbf{0}$
- ☐ Conclude $\dim V_j = 2j + 1$ and weights are $\{-2j, -2j + 2, \dots, 2j - 2, 2j\}$
- ☐ Normalize to get the standard ladder formulas with square-root factors
- ☐ Prove uniqueness: any irrep of dimension $2j + 1$ is equivalent to π_j
- ☐ Classification table: $j = 0$ scalar, $\frac{1}{2}$ spinor, 1 vector, $\frac{3}{2}, \dots$

Sat 25 Jul — Day 6: Integration to SU(2)

- ☐ Prove $\mathrm{SU}(2) \cong \mathbb{S}^3$ (homeomorphic to the 3-sphere)
- ☐ Conclude $\mathrm{SU}(2)$ is simply connected
- ☐ Theorem: every Lie algebra rep $\pi : \mathfrak{su}(2) \rightarrow \mathfrak{gl}(V)$ integrates uniquely to a Lie group rep $\Pi : \mathrm{SU}(2) \rightarrow \mathrm{GL}(V)$ via $\Pi(e^{\mathbf{X}}) = e^{\pi(\mathbf{X})}$
- ☐ Apply to π_j to get $\Pi_j : \mathrm{SU}(2) \rightarrow \mathrm{GL}(V_j)$

Sun 26 Jul — Day 7: Projective Representations & Descent

- ☐ Define projective representation: $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$
- ☐ Bargmann's theorem / physical motivation: in QM, symmetries need only be represented up to phase
- ☐ Prove: every projective representation of $\mathrm{SO}(3)$ lifts to an ordinary representation of $\mathrm{SU}(2)$
- ☐ Prove descent criterion: $\Pi_j(-\mathbf{I}_2) = e^{2\pi \cdot \pi_j(\mathbf{T}_3)} = (-1)^{2j} I_{V_j}$
- ☐ Conclude: integer j descends to ordinary $\mathrm{SO}(3)$ rep; half-integer j gives projective $\mathrm{SO}(3)$ rep (spinor)

Mon 27 Jul — Day 8: Physical Spin Operators

- ☐ Define Hermitian spin observables: $S_k = i\hbar \pi_j(\mathbf{T}_k)$
- ☐ Prove angular momentum algebra: $[S_i, S_j] = i\hbar \varepsilon_{ijk} S_k$
- ☐ Prove $S^2 = \hbar^2 j(j + 1)I$ and S_z has eigenvalues $\hbar m$ ($m = -j, \dots, j$)
- ☐ Example: $j = \frac{1}{2}$ — Pauli matrices $S_k = \frac{\hbar}{2} \sigma_k$ with explicit matrix forms
- ☐ Spinor sign change: rotation by 2π maps $|\psi\rangle \mapsto -|\psi\rangle$; restored after 4π

Tue 28 Jul — Day 9: Polish & Submit

- ☐ Proof-read all new content for mathematical correctness
- ☐ Ensure consistent notation with earlier sections
- ☐ Add cross-references (`\ref`) throughout new sections
- ☐ Run full compile with `latexmk` — zero errors, zero warnings
- ☐ Update abstract if needed to reflect completed content
- ☐ Final `./publish` to push to GitHub

Wed 29 Jul — Buffer Day

- ☐ Catch up on any unfinished items from Days 1–9

- ☐ Begin preparing presentation slides / talk outline

Thu 30 Jul — Final Review & Rehearsal

- ☐ Final paper read-through on paper
- ☐ Rehearse presentation with poster
- ☐ Verify QR code link works, PDFs are accessible on GitHub

Fri 31 Jul — PRESENTATION DAY

Feasibility: 9 working days + 2 buffer/prep days.

- Days 1–3 (Mon–Wed) are straightforward definitions + short proofs — **easy**.
 - Days 4–5 (Thu–Fri) classification theorem is the hardest — **2 focused sessions**.
 - Days 6–8 (Sat–Mon) are moderate — standard Lie theory, weekend coverage.
 - Day 9 (Tue) is polish — **light work**.
 - Wed 29 Jul — buffer for any spillover.
 - Thu 30 Jul — final review + presentation rehearsal.
 - **Total: ~17 new items**, most short. Your writing speed suggests this is well within range.
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